Figure 2: Two example ideas used as the basis for comparison in subsequent figures, evaluated by Idea Novelty Checker (Ours), AI Scientist, and AI Researcher.

Example 1

Idea: Develop a system that uses a *faceted author representation* of digital learning resource (DLR) creators based on their educational materials and inferred teaching personas. This system would aim *to support ubiquitous learning* by helping learners discover novel educators and materials that offer innovative perspectives. *Usability testing of learning resources* would be conducted to ensure the system enhances the learning experience by balancing relevance and novelty, thus boosting the accessibility and discoverability of diverse educational content.

Most Relevant Papers:

(1) Bursting Scientific Filter Bubbles: Boosting Innovation via Novel Author Discovery
(2) Bridger: Toward Bursting Scientific Filter Bubbles and Boosting Innovation via Novel Author Discovery
(3) Novel Algorithmic Recommendation Engine for Diverse Content Discovery
(4) ComLittee: Literature Discovery with Personal Elected Author Committees
(5) Explanations in Open User Models for Personalized Information Exploration
(6) AMiner: Mining Deep Knowledge from Big Scholar Data
(7) Similar researcher search in academic environments
(8) VeTo-web: A Recommendation Tool for the Expansion of Sets of Scholars
(9) From Who You Know to What You Read: Augmenting Scientific Recommendations with Implicit Social Networks
(10) DiscipLink: Unfolding Interdisciplinary Information Seeking Process via Human-AI Co-Exploration

Expert-Labeled Class: Novel

Example 2

Idea: Develop a **Co-Creative Interaction Framework** for LLM-assisted evaluations to **align llm evaluation with human preferences**. This framework will map out detailed interaction models between human evaluators and LLMs, including turn-taking, communication protocols, and iterative feedback loops. The framework will be evaluated through **qualitative study** involving user testing and thematic analysis of evaluator experiences, capturing how well the framework supports alignment with human preferences and identifying areas for iterative improvement.

Most Relevant Papers:

(1) EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria
(2) Human-Centered Design Recommendations for LLM-as-a-judge
(3) CheckEval: Robust Evaluation Framework using Large Language Model via Checklist
(4) Human-Centered Evaluation and Auditing of Language Models
(5) Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models
(6) Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences
(7) Evaluation of Code Generation for Simulating Participant Behavior in Experience Sampling Method by Iterative In-Context Learning of a Large Language Model
(8) Discovering Language Model Behaviors with Model-Written Evaluations
(9) LLM Comparator: Visual Analytics for Side-by-Side Evaluation of Large Language Models
(10) Do LLMs Meet the Needs of Software Tutorial Writers? Opportunities and Design Implications

Expert-Labeled Class: Novel

Figure 3: Reviews corresponding to idea 1 in Figure 2.

Method	Review
Scideator (Ours)	The idea is **novel** because it introduces the concept of faceted author representation specifically for digital learning resource (DLR) creators, focusing on their educational materials and inferred teaching personas to support ubiquitous learning. This approach is distinct from related works like Bridger[1][2], which focus on scientific author discovery, and ComLittee[4], which supports authorcentric exploration in research. The proposed system aims to enhance the discoverability of diverse educational content, balancing relevance and novelty, which is not explicitly addressed in the related papers.
AI Scientist	THOUGHT: The proposed research presents an innovative approach to language models. The rough goal is to train a model to predict the next character in a sequence in order to be able to capture the semantic meaning of the entire sequence, which can be used in language models. This approach is not uncommon, though, as there are many areas where we must generate predictions of the next likely character, which can be used in order to understand the entire meaning of the text. This model aims to improve upon traditional models by locally training the model on the use of characters rather than words. Of particular note is the fact that the model uses a scaled gradient descent method which changes how the weights are updated, a method that is very new and, thus, is a key point of innovation. While that is extremely novel, a model that works on the character level is not. The primary new technique here is the use of a momentum of the gradient, which is used to improve learning. The model uses a technique similar to many current models. It is possible that it could be used with different parameters, though, to learn more or to learn in a more effective way than other models. This approach has been suggested many times over the years, but it is promising to pursue. It is, however, a novel approach to the existing technology. However, we must consider the existing literature and how it might overlap with this proposition. I will now proceed to search for related publications that may be relevant in order to establish a definitive argument regarding the novelty of the proposed model. RESPONSE: “‘json\n\n "Query": "Character-level language models"\n\n“‘\n I believe this should identify any previous work or existing lines of research that are most similar to the proposed model. This will confirm whether this is actually a novel concept. My primary concern is understanding if this is a novel or an existing work, as this is a model that could potentially be used for many purposes. There is a good chance that it might have already been proposed. I hope that this search will bring more clarity to the argument. **If decision is not made, default is: not novel.**
AI Researcher	Proposed Research Problem and Approach: Project Proposal: The research problem is the limited accessibility and discoverability of innovative educational content due to the lack of effective representation and utilization of diverse teaching personas and educational materials of DLR creators. The approach involves developing a faceted author representation based on creators' educational materials and inferred teaching personas to enhance the discovery of novel educational content. Paper Abstract: The research problem is the isolation of scientific research and information overload, which limit awareness and hinder innovation. The approach involves constructing a faceted representation of authors from their papers and inferred personas to balance relevance and novelty in discovering new scholars and their work. Summary and Differences: - Project Proposal Summary: Enhancing the discovery of digital learning resources by developing a faceted author representation based on educational materials and teaching personas. - Paper Abstract Summary: Facilitating the discovery of new scholars and their work by constructing a faceted author representation from scientific papers and inferred personas to balance relevance and novelty. - Differences: - The project proposal focuses on digital learning resources and educational content, aiming to improve the learning experience. - The paper abstract focuses on scientific research and scholars, aiming to boost innovation and bridge diverse scientific communities. No .... *The comparison with other papers has been truncated due to space limitations. All papers in the most relevant set are marked as "Binary Judgment: No", which implies it* ***should not be cited as similar work to the idea.*** **Classification: novel**

Figure 4: Reviews corresponding to idea 2 in Figure 2.

Method	Review
Scideator (Ours)	The idea is **novel** because it introduces a CoCreative Interaction Framework specifically designed to align LLM evaluation with human preferences through detailed interaction models, including turntaking, communication protocols, and iterative feedback loops. This approach is distinct from related works like EvalLM[1], which focuses on iterative refinement of prompts, and EvaluLLM[2], which leverages LLMs as customizable judges but does not map out detailed interaction models. Additionally, CheckEval[3] and Prometheus 2[5] focus on evaluation frameworks and opensource evaluators, respectively, without the cocreative interaction aspect. The proposed framework's emphasis on qualitative studies and thematic analysis to capture evaluator experiences further differentiates it from existing methods.
AI Scientist	THOUGHT:\n The proposed idea involves designing a framework for co-creative interaction between human evaluators and Large Language Models (LLMs), focusing on aspects like turn-taking and communication protocols. It also emphasizes qualitative study and thematic analysis. This is a very specific angle that combines human-computer interaction and AI evaluation. While several existing papers discuss LLM evaluation frameworks, human-centered design, and human involvement in evaluation (e.g., EvalLM, EvaluLLM, and other papers from the results), the specific focus on co-creative interaction frameworks with detailed interaction models and iterative feedback loops may not have been explored thoroughly.\n\n Additionally, the idea looks at aligning evaluations with human preferences which is a nuanced approach. Therefore, a more targeted query to ascertain if any existing literature discusses similar co-creative frameworks or detailed interaction models involving human evaluators and LLMs is necessary. The current results do not seem to cover this specific angle in depth. **Decision made: novel** RESPONSE:\n"'json\n\n "Query": "Co-creative interaction framework LLM evaluation"\n\n"'
AI Researcher	Proposed Research Problem and Approach: - Project Proposal: - Problem: Current LLM evaluation methods often fail to align with human preferences. - Approach: Develop a Co-Creative Interaction Framework (CCIF) that includes detailed interaction models between human evaluators and LLMs, facilitating turn-taking, communication protocols, and iterative feedback loops to better align LLM evaluations with human preferences. - Paper Abstract: - Problem: Developers must iteratively revise prompts by evaluating outputs to diagnose weaknesses, which is effort-intensive and context-specific. - Approach: Present EvalLM, an interactive system for iteratively refining prompts by evaluating multiple outputs on user-defined criteria, using an LLM-based evaluator to provide feedback and improve prompts. - Differences: - The project proposal focuses on aligning LLM evaluations with human preferences through a co-creative interaction framework involving detailed interaction models. - The paper abstract focuses on refining LLM prompts through an interactive system that evaluates outputs based on user-defined criteria. No .... *The comparison with other papers has been truncated due to space limitations. All papers in the most relevant set are marked as "Binary Judgment: No", which implies it **should not be cited as similar work to the idea.*** **Classification: novel**